Name: **UIN:**

Graphene and 2D Materials: Science, Processing, and Applications (ChE 494)

Quiz 2: Processing of 2D Materials

October 22, 2019 **40 Minutes** **Total: 100**

1. Which of the following techniques is a top-down approach for the graphene synthesis?

   (A) CVD (B) Micromechanical Cleavage of HOPG (c) Epitaxial Growth on SiC

2. The Hummers method is used for producing graphene by oxidizing graphite to graphene oxide (GO) by using suitable oxidizing agents.

   (A) True (B) False

3. The growth of epitaxial graphene by sublimation method: Annealing the SiC crystal at 300 K.

   (A) True (B) False

4. Crystal Structure = Lattice + Basis (Example: Graphene Crystal Structure = Hexagonal Lattice + Basis).

   (B) True (B) False

5. A crystalline solid is the solid form of a substance in which the atoms or molecules are arranged in a random pattern.

   (A) True (B) False

6. Which of the following transition metals has highest carbon affinity?

   (A) Cu (B) Fe (C) Ni (D) Co

7. Graphene growth on Copper (Cu) is based on carbon segregation and precipitation process.

   (A) True (B) False

8. Transition metals and carbon solubility (Arrange with 'increasing' carbon solubility).

   (A) Fe-Co-Ni-Cu (B) Cu-Fe-Ni-Co (C) Cu-Ni-Co-Fe (D) Ni-Cu-Fe-Co

9. During the cooling down process, carbon atoms diffuse out from the NiC solid solution and precipitate on the Ni surface to form graphene films.

   (A) True (B) False

10. Which of the following process required to convert 'graphene oxide' to 'reduced graphene oxide'?

   (A) Sonication (B) Oxidation (C) Reduction

11. The graphene growth on Cu is a surface reaction process, which is not self-limiting.

    (A) True (B) False

12. Transition metals have partially-filled …………………. that enable them to adsorb and interact with the hydrocarbons that are used as the carbon sources for the synthesis of graphene.

    (A) s-orbitals (B) p-orbitals (C) d-orbitals

13. The graphene synthesis *via* chemical exfoliation route follows: graphite -> graphene oxide -> graphite oxide -> reduced graphene oxide.

    (A) True (B) False

14. A phase diagram is a graphical representation of the combinations of temperature, pressure, composition, or other variables for which specific phases exist at equilibrium.

    (A) True (B) False

15. The production of graphene on transition metals (Cu or Ni) *via* CVD requires transfer onto a desired substrate after etching off the metals for further characterizations and applications of graphene.

    (A) True (B) False

16. For the synthesis of 'white graphene' *via* CVD, the crystalline solid precursor (ammonia borane) is thermally decomposed to hydrogen, borazine, and amminoborane at temperature of 60∼180 °C.

    (A) True (B) False

17. Chemical vapor deposition (CVD) is a process where one or more gaseous species react on a solid surface and one of the reaction products is a …………………….. phase material.

    (A) Gas (B) Liquid (C) Solid

18. In case of graphene synthesis on metal catalyst substrates *via* CVD, graphene crystallites will seed at positions of locally 'lower chemical potential' such as: ………………………….

    (A) Grain Boundary (B) Dislocation and Defects (C) Contaminants (D) All

19. A polycrystalline solid is made up of an aggregate of many small single crystals.

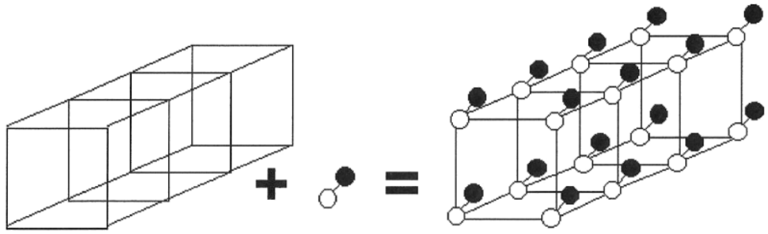
    (A) True (B) False

20. In case of epitaxial growth of graphene on SiC, the silicon atoms undergo sublimation from the SiC causes a carbon rich surface that nucleates an epitaxial graphene layer.

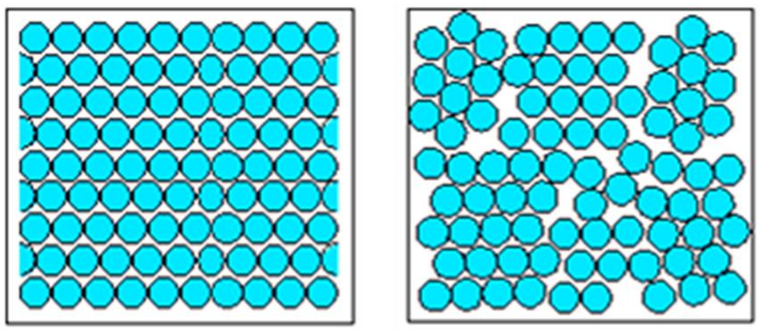
    (A) True (B) False

Name: **UIN:**

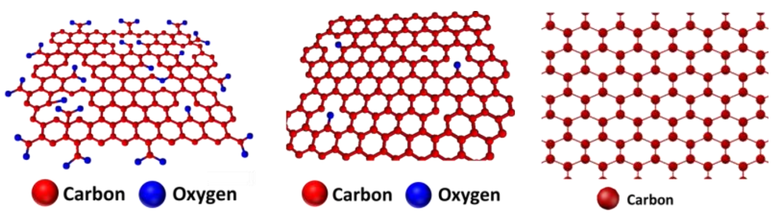
21. Designate lattice, basis, and crystal structure in the following figure.



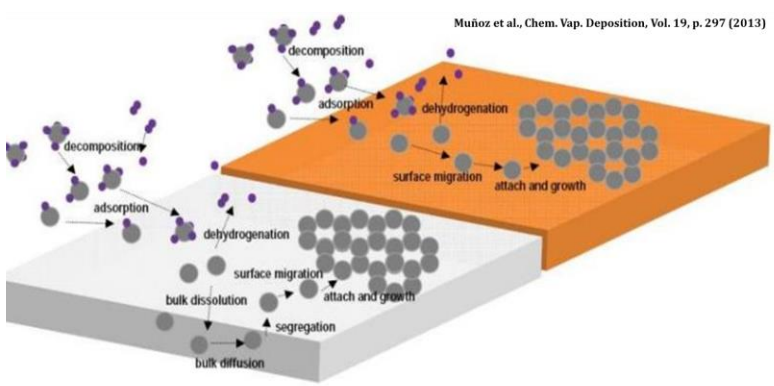
22. Differentiate between single crystal and polycrystalline solids from the following figure.



23. From the following figure: Highlight graphene, graphene oxide and reduced graphene oxide.



24. From the following figure: Differentiate between high and low carbon solubility metal catalyst.



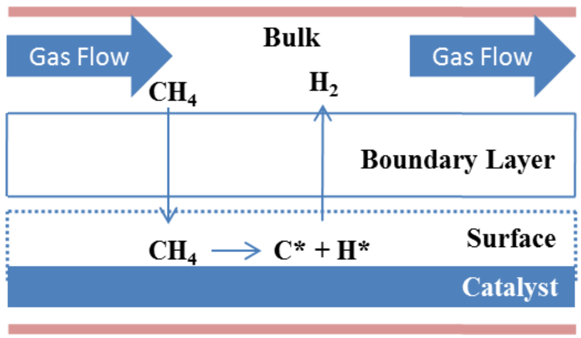
25. Which of the following sacrificial polymers can be used to transfer graphene from metal substrates.

(A) PMMA (B) PDMS (C) CAB (D) PET (E) All

Name: **UIN:**

Optional Question(s):

26. Consider the synthesis of high quality graphene on a catalytic transition metal surface with electronic configuration ([Ar] $3d^{10} 4s^1$). As shown in the following figure, there are two fluxes of the active species that coexist: (i) flux of active species through the boundary layer (mass transport region) and (ii) the rate at which the active species are consumed at the surface of the catalyst (surface reaction region) to form the graphene lattice. State the role of kinetic factors on the synthesis of graphene.



Answer: